



Experimental tests of fire resprouting in Hawaiian plants

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ABSTRACT

Context. Fire is increasing in intensity and frequency, leading to novel fire regimes in many regions. For oceanic islands of volcanic origin, endemic plants have evolved in the context of lava-ignited fires. **Aims.** Responses to fire were investigated in four endemic (*Nototrichium humile*, *Hibiscus brackenridgei* subsp. *mokuleianus*, *Gossypium tomentosum*, and *Erythrina sandwicensis*) and one invasive (*Leucaena leucocephala*) plant species in Hawai'i. Because conservation efforts commonly rely on juvenile stages, which are predicted to be particularly vulnerable to fire, we tested fire resprouting in saplings. **Methods.** Sapling resprouting was tested using experimental methods. Saplings were burned using a fire torch, and resprouting was monitored for 30 days. Proportion of plants resprouting and the timing of resprouting were the metrics of fire tolerance. **Key results.** Resprouting rates were high in four of the species (65–95%). The endemic *E. sandwicensis* was the only species that failed to resprout. The timing of resprouting varied among species, with the invasive species resprouting faster than the endemics. **Conclusions.** Saplings demonstrated surprisingly high tolerance to the fire treatments, indicating potential for these early stages to withstand fire. The invasive species resprouted faster than the endemics, which could contribute to competitive displacement in burned sites. At least one native Hawaiian plant species potentially lacks fire tolerance completely at the sapling stage. **Implications.** Sapling fire tolerance informs restoration actions that rely on out-plantings into fire-prone sites. Scaling up these results requires additional species testing and fire treatments to mimic contemporary fires.

Keywords: endangered species, fire tolerance, invasive species, island biodiversity, island plant syndromes, novel fire regime, ontogeny, resprouting rate.

Introduction

Fires have played a natural and integral part in shaping both the historical and modern structure and function of ecosystems worldwide (Bowman *et al.* 2009; Pausas and Keeley 2009). However, since the late 18th century, anthropogenic factors have drastically altered global fire regimes and climate through industrialization and landscape modification, and interfered with Indigenous fire management practices (Bowman *et al.* 2020). Although fire regimes vary tremendously across regions (Earl and Simmonds 2018), novel anthropogenic fire regimes are generally characterized by increasing frequencies and intensities of fire, including more extreme, catastrophic fires (Ellis *et al.* 2022; Jones *et al.* 2022). In regions with historically low-severity or infrequent fires, plants may be poorly adapted to modern fire regimes, identifying fire as a major threat to biodiversity and ecosystem function (Hu *et al.* 2015; Kelly *et al.* 2020).

Plants have evolved diverse strategies to tolerate damage from fire, allowing them to thrive in fire-prone habitats (Keeley *et al.* 2011). The two primary fire tolerance strategies are: (1) seeding; and (2) resprouting, which have evolved convergently across diverse phylogenetic lineages under different fire regimes (Bellingham and Sparrow 2000; Lamont *et al.* 2019). The seeding strategy typically involves fire stimulated flowering, seed production, or germination, and so is likely to be a derived trait that evolved in habitats with predictable fires of relatively high intensity, corresponding to the production of

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sites favoring seedling establishment (Lamont *et al.* 2019). In contrast, the resprouting strategy is thought to be an ancient trait, associated with tolerance to diverse disturbances, including fire, as well as drought, herbivory, storms and landslides, and freezing damage (Bond and Midgley 2001; Pausas and Keeley 2014). Fire tolerance via resprouting is thus likely to be more widespread, and even found in plants that evolved under historically infrequent or low-intensity fires because of its generalized role in disturbance tolerance (Keeley *et al.* 2011). Because resprouting depends on an available bud-bank and stored nonstructural carbohydrates to support new growth, it is predicted that resprouting capacity is low in young plants, increasing quickly during the juvenile stage, followed by a period of high capacity through maturity until senescence leads to a gradual loss of resprouting (Clarke *et al.* 2013). Mixed evidence supports this ontogeny-related prediction for resprouting, with variable patterns reported across studies (Bond and Midgley 2001; Del Tredici 2001; Keeley 2006; Shibata *et al.* 2014), likely reflecting non-linearity across ontogenetic stages and different stages compared across studies. Nonetheless, it is clear that resprouting capacity changes across the lifespans of plants, identifying this as a potentially important, but understudied, factor contributing to the resilience of plant populations to changing fire regimes.

Oceanic islands are experiencing dramatic shifts in fire regimes, as a consequence of human activities, land use change, and the accumulation of highly flammable invasive species (Trauernicht *et al.* 2015; Trauernicht *et al.* 2024). For many islands, fire was historically infrequent or of low severity as a consequence of limited land area for large wildfires to develop and because lightning strikes are relatively less common, particularly on small, remote islands (Albrecht *et al.* 2016). Thus, it has been predicted that a feature of the island plant syndrome is weak fire tolerance (Burns 2019). However, for volcanic islands, fire is a natural and common part of the disturbance regime, causing widespread destruction to island plants and ecosystems (Nogales *et al.* 2022). Indeed, fire tolerance via resprouting and seeding has been described for several native plant species from volcanic islands (Arévalo *et al.* 2001; Ainsworth and Kauffman 2009; Teixeira *et al.* 2020). Furthermore, other disturbances are common to islands, such as landslides, tropical storms, herbivore outbreaks, and drought (Fernández-Palacios *et al.* 2021), which may have selected for a generalized disturbance tolerance strategy in island plants (Barton *et al.* 2025). As islands account for a disproportionate extent of global biodiversity as a consequence of high rates of endemism (Fernández-Palacios *et al.* 2021), investigating the potential resilience of native island plants to fire is an urgent conservation priority. Because fire generally coincides with other global threats on islands, it is challenging to precisely infer fire tolerance under field conditions. Experimental methods that quantify fire tolerance directly, while controlling for other confounding

stressors, offer the most precise approach to test fire tolerance in island plants.

The Hawaiian Islands offer a compelling system for investigating fire tolerance in endemic island plants. Although little is known about the history of wildfires in the Hawaiian Islands prior to human contact (Smith and Tunison 1992), it is generally assumed that wildfires commonly occurred during volcanic activity, but that these rarely spread far or reached very hot temperatures due to the lack of fuel associated with low flammability of the native vegetation (Walker and Bellingham 2011). Fire regimes changed after the settlement of early Hawaiians. As a consequence of land clearing and the expansion of pili grass (*Heteropogon contortus*) used for thatching material, this has resulted in an increase in fire frequency across the archipelago (Kirch 1982). Land use further changed rapidly after the arrival of Europeans and Americans, as agriculture and ranching lands sprawled, aiding in the spread of fast-growing, fire-adapted grass species introduced to support agribusiness (Trauernicht *et al.* 2015). In the past 100 years alone, the average area of land burned annually has increased by 400%, with continued increases expected from the statewide abandonment of sugarcane and other fallow agricultural lands (Trauernicht *et al.* 2015). With exceptionally high rates of endemism, >90% for angiosperms (Price and Wagner 2018), the Hawaiian flora is globally unique. There is widespread evidence of native plant declines across the islands, attributed at least in part to anthropogenic fire (D'Antonio *et al.* 2011; Barton *et al.* 2021).

Native plant declines coincide with increased fire frequency and intensity across the Hawaiian Islands (Jacobi *et al.* 2017), suggesting weak fire tolerance, and consistent with the island plant syndrome prediction (Burns 2019). Yet, some studies have reported high rates of resprouting and seeding in native plants following volcanic fire on Hawai'i Island (Ainsworth and Kauffman 2009; Loh *et al.* 2009). It is likely that fire tolerance has evolved in response to volcanic activity in at least some Hawaiian plants, although simultaneous stress from fire and competition with invasive grasses may limit the extent that fire tolerance traits translate into persistence and population stability under contemporary global change (Tunison *et al.* 2000; Ainsworth and Kauffman 2010). Moreover, in degraded, recently burned habitats where restoration efforts are actively supplementing native plant populations through out-planting (Trauernicht *et al.* 2018; Warneke *et al.* 2023), low fire tolerance in juvenile plants may limit success under modern fire regimes. Previous studies in Hawai'i have focused exclusively on resprouting in adult plants and regeneration via seed following fire, highlighting juvenile plant fire tolerance as an important knowledge gap, particularly in light of restoration activities that rely on juvenile out-planting survival for success.

In this study, we address this knowledge gap and investigate resprouting following fire in saplings using an experimental approach. Our species pool includes five woody species that occur in low-elevation aridlands where fire has

increased in frequency and intensity under the modern fire regime (Table 1). Four endemic shrubs and trees that occur in high fire-risk zones were tested, including two endangered species that are threatened by fire: (1) kulu'i (*Nototrichium humile*); and (2) ma'o hau hele (*Hibiscus brackenridgei* subsp. *mokuleianus*) (IUCN 2023). Two previously common species that are now in decline were also included: (1) ma'o (*Gossypium tomentosum*); and (2) wiliwili (*Erythrina sandwicensis*). In addition, we tested one highly invasive tree/shrub, koa haole (*Leucaena leucocephala*), which is common in aridlands where fire is frequent. We predict generally low resprouting rates, as expected for juvenile plants (Clarke *et al.* 2013). However, we expect that the invasive koa haole will have higher resprouting than the island endemics as it has demonstrated fire tolerance in other invasive populations (Santiago-García *et al.* 2008; Da Luz *et al.* 2024), including fire survival at the juvenile sapling stage (Wolfe and Van Bloem 2012). This study represents the first examination of juvenile plant responses to experimental fire in the native Hawaiian flora, offering new insights into the evolution of fire tolerance in island endemics and providing critical information for their conservation.

Methods

Study system

We tested fire tolerance using an experimental approach at the University of Hawai'i at Mānoa Pope Greenhouses. Plants used in the experiments were grown from wild-collected seeds or cuttings of cultivated plants (the endangered species) from O'ahu Island. Seedlings were grown in pots (approximately 3.7 L in volume) filled with equal parts by volume sunshine mix #4 (Canadian Sphagnum peat moss, coarse perlite, dolomite lime) and coarse perlite, and plants were watered daily. After plants had produced at least one lateral branch and most had reached a height of at least 20 cm, they were prepared for the fire treatment.

Fire experiments

Within each species, 20–23 replicate plants were treated (Table 1), and all replicates within a species were burned on a single day in 2020–2021 to minimize variability in climate conditions during the burn treatment. The five species could not be treated on the same day due to variability in growth rates and the availability of greenhouse space. All tested plants were juvenile stage saplings, but because of variability in size, stem diameter (cm) was included as a predictor variable in the statistical models to investigate if size influenced resprouting. Plants were moved outside of the greenhouse for the experimental burns, which were conducted on days with low wind velocity (<16 km per h) to minimize disruption of the flame during treatments. Mean temperatures in Mānoa ranged from 20.9 to 25.4°C (Longman *et al.* 2024).

Plants were experimentally burned using a prong-shaped fire torch, consisting of a stainless steel pipe assembled with two arms in a 'U' shape, venturi pipe, pressure regulator, and propane tank (Fig. 1), modeled after similar tools used for experimental fire research (Kral *et al.* 2015). Plants were burned individually by placing the plant between the torch arms at a suitable level to position the flame at the lower plant stem (Fig. 1). Plants were burned for a duration of 15, 30, or 60 s, during which time, leaves fully burned, and stems turned black. Due to limited sample sizes, the three burn durations could not be applied to all species (Table 2). Instead, at least two burn durations were randomly assigned to each species, with different durations assessed in different species (Table 2). For a single species (ma'o), all three burn durations were tested, but this compromised replication within treatment groups and so was not applied to the other species. To monitor potential temperature variability between treatments, thermocouple sensors were placed within the flame range, and temperatures were recorded on a data logger at 1-s intervals for the duration of treatments (Kral *et al.* 2015). This method produced burn heat curves (Fig. 2) that resemble the heat temperature and duration of prescribed grassland fires (Kral *et al.* 2015). Burned plants were returned

Table 1. Summary of five focal species included in fire tolerance experiments.

Species	Family	Habit	Origin (threatened)	Total % resprouting (n)	Day of first resprouting	Median resprouting day	Basal/epicormic sprouts (n)
<i>Nototrichium humile</i> (kulu'i)	Amaranthaceae	Shrub	Endemic (EN)	95% (20)	17	17	0/19
<i>Hibiscus brackenridgei</i> (ma'o hau hele)	Malvaceae	Shrub	Endemic (CR)	78% (23)	18	21	0/18
<i>Gossypium tomentosum</i> (ma'o)	Malvaceae	Shrub	Endemic (VU)	65% (20)	7	17	3/10
<i>Erythrina sandwicensis</i> (wiliwili)	Fabaceae	Tree	Endemic (VU)	0% (20)	–	–	–
<i>Leucaena leucocephala</i> (koa haole)	Fabaceae	Tree	Non-native	70% (20)	5	7	0/14

Four endemic species were included, as well as a single non-native species. Threat status as reported by the IUCN Red List (accessed 19 May 2024) is reported as critically endangered (CR), endangered (EN), or vulnerable (VU). Experimental results are reported for total % saplings that resprouted during the recovery period (n = species sample sizes), the first day a sapling was observed to resprout, the median day for resprouting, and the total number of saplings that showed basal (near ground) and epicormic (from stem) resprouting.



Fig. 1. Experimental fire torch, consisting of a stainless-steel pipe assembled with two arms in a 'U' shape, venturi pipe, pressure regulator, and propane tank powered by propane tank. (a) Experimental burn showing full torch and plant stand. (b) Wiliwili sapling (*Erythrina sandwicensis*) before burn treatment. (c) Close-up of wiliwili sapling undergoing experimental burning.

Table 2. Resprouting frequencies of the five focal species under three burn duration treatments (15, 30, 60 s).

Species	15-s burn	30-s burn	60-s burn
<i>Nototrichium humile</i> (kulu'i)	100% (10)	90% (10)	–
<i>Hibiscus brackenridgei</i> (ma'o hau hele)	91% (11)	67% (12)	–
<i>Gossypium tomentosum</i> (ma'o)	80% (10)	71% (7)	0% (3)
<i>Erythrina sandwicensis</i> (wiliwili)	–	0% (10)	0% (10)
<i>Leucaena leucocephala</i> (koa haole)	–	100% (10)	40% (10)

Percentage resprouting is reported, with sample sizes in parentheses. Not all species were tested across the three durations. Dash indicates missing treatments.

to the greenhouse, watered daily, and inspected every 2 days for 30 days to assess resprouting. Position of resprouting was recorded as epicormic (aerial, from stem buds) or basal (at or

below ground level, often involving belowground buds) (Clarke *et al.* 2013).

Data analysis

Resprouting response and timing were analyzed using generalized linear models with the glm function in the software environment R ver. 4.1.1 (R Core Team 2021) and RStudio ver. 2021.09.0 (RStudio Team 2021). Species were first analyzed in global models to test for variability among species in resprouting (yes/no) and resprouting day. In addition, because species were not treated with the same burn durations, they were also analyzed with individual models. In all cases, the models include two fixed factors: (1) burn duration (15, 30, 60 s, where appropriate); and (2) plant stem diameter (cm) measured at the time of the fire treatment.

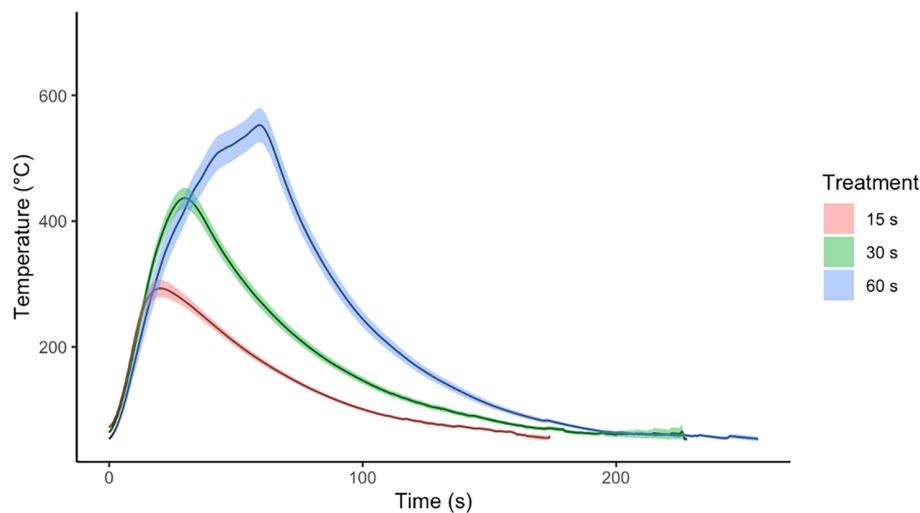


Fig. 2. Mean temperature curves (across all species and trials) for the three burn durations (15, 30, and 60 s).

Resprouting response is a binary variable (yes/no) and was fitted on a binomial distribution using the logit link. No overdispersion was detected for any species for the resprouting models (dispersion ratios > 1.08 , $P > 0.810$). Resprouting day is continuous and fitted with Poisson distribution. For ma'o alone, overdispersion was detected (dispersion ratio = 4.325, Pearson's $\chi^2 = 43.25$, $P < 0.001$). Switching to a negative binomial model (glm.nb in R) adequately accounted for extra-Poisson variation, with no overdispersion detected (dispersion ratio = 1.164, $P = 0.608$). For the other species, there was no evidence for overdispersion for the resprouting day models using the Poisson distribution (dispersion ratios > 0.63 , $P > 0.851$). The interaction between diameter and burn duration was tested in the individual species models and was removed from models in which it did not explain a significant extent of variance. Diameter data were not collected for wiliwili; therefore, this species was analyzed only with respect to burn duration. Test statistics are reported as χ^2 and F -values (Type III) for resprouting response and day, respectively.

Study permits

Our study was covered by: (1) US Fish and Wildlife Recovery Permit ES043638; and (2) Natural Area Reserve, Rare Plant, and Native Invertebrate Research Permit, Hawai'i Department of Land and Natural Resources, Endorsement Number I5457.

Results

In response to experimental burning, sapling resprouting rates were high ($>65\%$) for four of the five species tested (Table 1). Wiliwili was the only species with no observed resprouting, suggesting that the fire treatments killed these plants.

Kulu'i had the highest level of resprouting (95%), with all but one individual resprouting after fire treatment, followed by ma'o hau hele (78%). The global model revealed significant variability among species ($\chi^2_4 = 41.705$, $P < 0.0001$), and burn duration also significantly influenced resprouting ($\chi^2_1 = 19.43$, $P < 0.0001$). Plant size (i.e. stem diameter) was not included in the global model because it was not measured for wiliwili. In the individual species models, burn duration significantly influenced resprouting in ma'o and koa hoale (Table 3), with lower rates of resprouting following longer fire treatments (Table 2), presumably as a consequence of the much higher temperatures reached by longer burns (Fig. 2). Plant stem diameter significantly influenced resprouting response only in the invasive koa hoale; saplings that resprouted tended to be smaller (mean diameter 0.564 ± 0.063 cm) than saplings that did not resprout (mean diameter 0.620 ± 0.045 cm).

The global model for resprouting day indicated significant variability among species ($F_3 = 23.38$, $P < 0.0001$). Despite having the highest rates of resprouting, the two endangered species were substantially slower to resprout, with the first sign of resprouting on Days 17–18, 12 days later than resprouting was observed in the invasive koa haole. Among the four species that resprouted, the Hawaiian cotton, ma'o was the fastest native species to resprout (7 days after burn treatments), although it had the lowest resprouting rate (65%) overall. The global model further detected a significant effect of burn duration ($F_1 = 22.45$, $P < 0.0001$), but not stem diameter ($F_1 = 3.45$, $P = 0.1471$) on resprouting day. From the individual species models, koa haole demonstrated the longest range in the timing of resprouting, and this was significantly influenced by both burn duration and plant size (Table 3). Moreover, we detected a significant interaction between duration and size on resprouting day in koa haole (Table 3; see Supplementary Fig. S1), but not in any of the other species (interaction

Table 3. Results from generalized linear models testing effects of burn duration (15, 30, 60 s) and plant size (stem diameter, cm) on resprouting of four species. Bolded values indicate significant difference at $P < 0.05$.

Species	Resprouting (yes, no)			Resprouting day			
	<i>n</i>	Burn duration	Stem diameter	<i>n</i>	Burn duration	Stem diameter	Duration × diameter
<i>Nototrichium humile</i> (kulu'i)	20	1.43 (0.2303)	0 (1.0)	19	–	–	–
<i>Hibiscus brackenridgei</i> (ma'o hau hele)	23	2.20 (0.1376)	1.19 (0.2902)	18	2.58 (0.1294)	0 (0.9792)	–
<i>Gossypium tomentosum</i> (ma'o)	20	6.12 (0.0133)	0.37 (0.5419)	13	0.05 (0.8300)	2.21 (0.1682)	–
<i>Leucaena leucocephala</i> (koa haole)	20	11.01 (<0.0001)	4.73 (0.0297)	14	12.56 (0.0053)	11.09 (0.0076)	5.99 (0.0007)

Resprouting is a binary response variable, and is tested with a chi-square distribution; d.f. = 1. The day that plants resprouted was analyzed on a Poisson distribution or fitted with a negative binomial model (ma'o) with a Type III F -test statistic. Test statistics (χ^2 or F for resprouting and day, respectively) are reported, with P -values in parentheses. Models were not run for *E. sandwicensis*, which failed to resprout, or for day for *N. humile*, which failed to converge because of synchronous resprouting 17 days after the burn treatment.

removed from the final models). While there is no detectable effect of plant size in resprouting day for koa haole plants exposed to the 30-s burn, smaller plants in the 60-s burn group had considerably faster resprouting than larger plants (Table 2, Fig. S1).

The vast majority of resprouting was from the above-ground stem (i.e. epicormic resprouting, Table 1). Ma'o was the only species to exhibit basal resprouting from belowground tissues, with three plants resprouting from basal buds. The onset of basal resprouting appears to be slower than epicormic resprouting in ma'o as this was observed on Days 23–26.

Discussion

For this experiment, we predicted low survival and resprouting in response to fire both because saplings are thought to be constrained by their architecture and physiology, lacking sufficient nonstructural carbohydrates and the bud-bank necessary to resprout (Clarke et al. 2013), and also because island plants are expected to have lost fire tolerance as a consequence of evolving in regimes characterized by infrequent, low-severity fires (Burns 2019). Surprisingly, in contrast to these predictions, we observed high resprouting rates in three of the four endemic species, as well as in the invasive species. Resprouting rates varied among species, from 65 to 95% of burned saplings resprouting following fire. These rates are similar to the resprouting rates reported for native Hawaiian tree ferns (*Cibotium glaucum*) and the forest-dominant, *Metrosideros polymorpha* (40–90%), following a natural lava-ignited fire on Hawai'i Island (Ainsworth and Kauffman 2009). Our sapling resprouting rates are even comparable to those of fire-adapted *Eucalyptus* juveniles, which ranged from 78 to 100% following an experimental field fire (Werner and Franklin 2010). The only species that failed to resprout is the endemic wiliwili (*E. sandwicensis*), corroborating a previous field study (Smith and Tunison 1992) and suggesting that this species is a potential outlier with very low tolerance, and thus high vulnerability to fire disturbance. The lag time following fire, before the plants

resprouted, was also highly variable among species, and this undoubtedly plays an important role in vegetation recovery dynamics following fire. Considering these results, it seems that native Hawaiian plants are declining under novel fire regimes, not because of an intrinsic lack of capacity to tolerate fire, but rather because of shifting interactions as a consequence of the severity of fire and the timing of resprouting after fire.

These results corroborate previous studies reporting fire tolerance in native Hawaiian plants in the adult stage. Most of the previous evidence for resprouting and seeding following fire comes from Hawai'i Volcanoes National Park (HAVO), where active volcanoes ignite wildfire on a regular basis. For example, among 22 woody plant species investigated, 16 species resprouted and 15 species established seedlings in burned sites (Ainsworth and Kauffman 2009). Seedling establishment following fire in HAVO has also been demonstrated using experimental methods via out-plantings and controlled burns (Loh et al. 2009). Native Hawaiian tree ferns have also demonstrated fire tolerance via resprouting and postfire seedling establishment (Ainsworth and Kauffman 2009). Fire tolerance is not surprising for species and populations established on Hawai'i Island near the currently active volcanoes of Mauna Loa and Kilauea because fire is an ongoing source of selection pressure. Fire tolerance has also been documented for native plants on other islands with active volcanoes, such as the Canary Islands (Arévalo et al. 2001), the Aeolian Islands (Zaia et al. 2020), and the Aleutian Islands (Talbot et al. 2010). Such widespread evidence for fire tolerance in island plants contradicts the island plant syndrome prediction, but this was expected given the association between fire and volcanic activity, and the numerous oceanic islands that have formed via hotspot volcanoes (Barton et al. 2025). Our results are the first indication of fire tolerance in plant populations from O'ahu Island where the main period of volcanism ended approximately 2 million years ago (Ozawa et al. 2005). The persistence of fire tolerance in plant populations that have not experienced volcano-ignited wildfire for tens of thousands of generations is striking, and could indicate that the capacity

to tolerate fire does not impose costs so that it is selectively neutral (and thus, not selected against in the absence of fire), that there is gene flow between islands, that resprouting and seeding are adaptive in response to other stressors (i.e. that they are exaptations), or that fire tolerance is now adaptive in the context of the modern anthropogenic fire regime (Kelly *et al.* 2025). A notable outlier in our study is wiliwili, which failed to resprout at all following fire. It is unlikely that this result pertains to the focal population from O'ahu because wiliwili has also been reported to lack resprouting following fire on Hawai'i Island (Smith and Tunison 1992). However, wiliwili was the only native species that was not treated at the shortest fire duration of 15 s. It may be that wiliwili saplings can tolerate fires of very short durations, which would require additional experimentation to confirm. Further research exploring more species and intraspecific variability in fire tolerance for widespread species across the Hawaiian Archipelago would shed further light on patterns across islands with variable fire histories and elucidate the sources of selection pressure.

Fire regimes associated with volcanic activity are highly variable, reflecting volcano behavior, fuel loads of nearby vegetation, as well as climate (Quah *et al.* 2023). In Hawai'i, volcano-ignited wildfires would historically not have spread far or reached very high temperatures as a consequence of limited fuel accumulating from a native understory dominated by ferns, bryophytes, and some forbs and bunchgrasses. However, invasion by fire-promoting grasses has altered the contemporary wildfire dynamic dramatically, with high fuel loads accumulating from dried grass material contributing to catastrophic fires that spread quickly and burn very hot (D'Antonio and Vitousek 1992). Along with increases in anthropogenic ignitions in recent decades, Hawai'i now has one of the highest per-area wildfire rates in the United States (Trauernicht *et al.* 2015). In the context of this novel fire regime and invasive plant competition, fire tolerance evolved to volcano-ignited fire regimes is likely to be insufficient to promote native plant persistence in the Anthropocene. Although the native species tested here clearly have bud banks that are capable of resprouting following fire, nearly all resprouts emerged epicormically from stems. Protection of stem buds from very hot temperatures requires thick bark to prevent fatal heat damage (Clarke *et al.* 2013). Endemic Hawaiian plants are unlikely to have evolved the thick bark necessary to protect epicormic buds, as evidenced by lower rates of resprouting in response to longer burns in our experiments. However, future research is needed to quantify bark thickness and to determine the fatal temperature thresholds for native plants under novel fire regimes to confirm whether grass-fueled wildfires exceed the resprouting thresholds.

Invasive grasses are a major threat to native Hawaiian plants, not just because of their effects on the fire dynamics, but also by altering the recruitment niche that native plants depend on for seedling regeneration. Invasive grass understories

reduce native seedling establishment across the Hawaiian Islands, presumably by reducing light and water availability (Cabin *et al.* 2002; Thaxton *et al.* 2012). Yet, some invasive woody plants successfully establish within grass-invaded pasture, including koa haole (*L. leucocephala*). Koa haole is now among the most abundant species in the Hawaiian Islands (Craven *et al.* 2018), particularly in lowland arid habitats where contemporary fire poses the greatest threats. *L. leucocephala* has also invaded many other tropical and subtropical regions (Sharma *et al.* 2022), including other islands such as Puerto Rico where it has also been documented to be fire tolerant (Santiago-García *et al.* 2008; Wolfe and Van Bloem 2012). In our experiments, koa haole demonstrated a similar rate of resprouting as the endemic species (aside from wiliwili), but it resprouted much faster. The first koa haole resprouted 5 days following fire, and by Day 7, half of the saplings had resprouted. Ma'o also had an early resprouter at Day 7, but did not reach 50% resprouting until Day 17. Demonstrating even slower resprouting, the two endangered species, kulu'i (*N. humile*) and ma'o hau hele (*H. brackenridgei*), first resprouted on Days 17 and 18, respectively. Such slow resprouting responses in the endemic species are likely to be a liability in the competitive dynamics following fire when invasive grasses and koa haole resprout and re-establish faster.

Conservation of native Hawaiian plants must thus consider not just novel fire disturbance, but also the subsequent species interactions that determine post-fire vegetation recovery. It is promising that we detected resprouting potential in such young plants in this experiment, which suggests some fire tolerance in juvenile plants. Fire tolerance during early ontogeny may promote regeneration and stable population dynamics in the context of fire (at least, volcanic-ignited fire), and it may inform conservation efforts. Seedling and juvenile stages feature prominently in conservation plans because they are often used in restoration. Out-planting seedling and juvenile plants is a common practice to augment existing populations or to re-establish native species in cleared or disturbed sites (Lázaro-González *et al.* 2023). Direct seeding, which ultimately depends on successful seedling establishment, is also commonly used in restoration (Larson *et al.* 2023). In Hawai'i, out-planting and seeding of native and Polynesian introduced species are part of most conservation plans, and although these often explicitly consider global threats such as invasive competition and fire (Trauernicht *et al.* 2018; Burnett *et al.* 2019), direct evidence of species-level tolerance is often missing. Experimental studies testing tolerance to stressors, such as fire, are needed to confidently predict plant responses to stressors, particularly at the early ontogenetic stages targeted for restoration. Scaling up from the few species tested here through additional experimentation, combined with field studies to translate plant responses detected under experimental conditions to those of field-grown plants facing additional stressors and interactions, is urgently needed to characterize broader patterns in fire tolerance via resprouting

in Hawai'i's native and invasive species to conserve this unique island flora in the face of novel fire regimes (Barton et al. 2025).

Supplementary material

Supplementary material is available online.

References

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